6.3 Temporal Static Visualisation of Transcripts for Pre-Analysis of Video Material: Identifying Modes of Information Sharing

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Abstract: When evaluating group performance in the field of group collaboration different forms of information sharing types are utilised as indicators for well performing or under performing groups. A transcript-file of a group conversation provide us with such communication patterns in form of the order of speaker turns and the amount of words assigned to each speaker at a particular instance. In this study we explore how one can utilise transcript meta data and the field of data visualisation to help a researcher to deduce insights for further investigation of the studied conversation faster. This would help qualitative researchers in the pre-analysis of bigger data sets. In Python we wrote a script that - from transcript meta-data - presents Temporal Static Visualisations (TSV) of the group members individual contribution as well as the mean-variance in participation over time. This supports the researcher in identifying *sequences of* and *transitions between* monologues, dialogues and group discussions in the observed conversation.

The main strength of TSV lies in the static inclusion of time development that helps the researcher to gain a quick insight into the flow of the observed conversation. However we still face the challenge of providing the design researcher with more contextual information. Thus, proposed future work should focus on how to automatically include additional context and content information such as artefact usage and physical interactions in the visualisations.

Keywords: pre-analysis of transcript data, temporal data visualisation, design observation, engineering design, group dynamics.

1. Introduction

For decades, video has been one of the main data sources for design research experiments, especially on group behaviour (Tang, 1991). Alas, coding transcript data is very time consuming in terms of educating coders as well as conducting the actual coding itself. Moreover, it can be difficult to keep an overview over the large amount of data. This paper proposes a new method utilising data visualisation of transcript meta-data to help researchers discover different modes of information sharing in such video material for further content analysis. We present an approach for pre-analysis of transcript data in relation to group dynamic research in the field of engineering design. By extracting meta-data from the transcript file we visualise the conversation contribution over time among the group members as well as plotting mean-variance of group members participation. We call the approach Temporal Static Visualisation (TSV). TSV allows the researcher to quickly identify different forms of information sharing types, as monologues, dialogues or group conversation as well as the transation between the types.

Our approach was initiated by the recent trend of dynamic data visualisation, which in recent years has created valuable aids for qualitative researchers. Tools as Leximancer, Gephi and

Cytoscape for example allow investigators to get new insights into bigger datasets (Bastian, Heymann, Jacomy, & others, 2009; Smith, 2003; Smoot, Ono, Ruscheinski, Wang, & Ideker, 2011). However most of the tools only provide the researcher with holistic overview of the data like semantic networks, but fails at helping the researcher deduce specific sequences for further investigation. One exception is Dong's (2005) analysis of semantic coherence that investigate the development of a common language within design teams (A. Dong, 2005).

In the field of audio visualisation several researchers have explored and evaluated the effect of live visualising dialogue flow in a group conversation (Avci & Aran, 2016; Bergstrom & Karahalios, 2007b; DiMicco, Hollenbach, & Bender, 2006). The approach is different compared to a design researchers focus since the main aim is to actively attempt to affect the discussion and speaking patterns by visualising the live flow of communication. There is a focus on preventing dominant participants to take over the discussion and having less talkative participants to express their opinions (Bergstrom & Karahalios, 2007a). The main assumption in the field of audio visualisation in group discussion is that one get a more diverse type of information sharing, when all group members contribute in the discussion. Hence the overall main hypothesis is that there are in fact sufficient and less sufficient dialogue patterns for different contexts. We use this hypothesise to claim that by looking at certain dialogue patterns one can identify interesting video passages of a group conversation in design work as well.

Hence, the research question of this paper has been:

How can design researchers apply data visualisation on group conversations for identification of different modes of information sharing?

First, we present related work. Next, we describe the development of a dynamic network visualisation approach. This we evaluated by interviewing reseachers of a research group at the Technical University of Denmark (DTU) also working with the same data set from DTRS11 (Christensen & Abildgaard, 2016). For further reference we will mention this research group as *the DTU group*. With learnings from this evaluation we finally present the suggested approach; Temporal Static Trancript Visualisation (TSV) where time perspective, conversation participation and content are captured in one graph. Finally, we discuss the results and limitations of TSV as well as highlighting opportunities for improving the approach further.

2 Related Work

Our approach are grounded in three different research fields; observational case studies of group dynamics in Design Research, Dynamic Data Visualisation and finally Investigation of Group Interactions through Visualiations of conversation contribution. Below the three fields will be elaborated.

2.1 Group Dynamic Analysis and Design Research

Throughout the history of design research video observations has served as an often used tool for case based observational studies of group collaboration (Tang, 1991). Sociologists approach group collaboration in the view of seeing it as a system of human interaction. At this degree of abstraction there is no necessary incongruity in comparing cases with each other since the categories of interaction are more general such as "showing empathy" or "ask for clarification". Robert F. Bales defined in 1950 a set of twelve categories for the analysis of small group

interaction (Bales, 1950). This set of categories has served as starting point for Bales himself and other socio psychologists developing more detailed sociological frameworks (Scanzoni, 1983; Sjovold, 2007). Bales' method is called the Interaction Process method and is based on the observation of behaviour act by act. The researcher codes every single act in the categories of the defined interactions such as: 'Shows solidarity', 'Shows tension release', 'Gives suggestion', 'Ask suggestion', 'Shows Tension', 'Shows Antagonism'. The coding mainly focuses on language, but can also take body language (e.g. nodding or geturing) into account. By giving scores for each act, the researcher will end up with an empirical score chart of a group interaction which is comparable with other group observations. This provides an insight in what type of group interaction has happened as well as defining role and team dynamics in the group. Sonnenwald (1996) focuses on the different communication roles in the design process rather than only behaviour. Through evaluating several different case studies of team work all involving the design process they define 13 different types of communication roles appearing in the conversation. Sonalkar, Mabogunje, & Leifer (2013) are focussing on design interactions in design work and introduce the Interaction Notation Dynamics, which seeks to cover the process of moment to moment concept development and capture the interpersonal interactions. By not only introducing 13 different types of interactions in a concept development dialogue, but also assigning them a specific symbol they provide the community with a standardised coding scheme for researchers to use. The 13 interactions are 'move', 'question', 'hesitation', 'block', 'support for move', 'support for block', 'overcoming', 'deflection', 'interruption', 'yes and', 'deviation and humour'.

We define the previous three mentioned cases as semi-quantitative since they all require an educated coder to process the empirical data. A more quantitative approach for analysing group interactions was introduced by Dong (2005) and is based on the development of a coherent language throughout a group discussion. The method is called Latent Semantic Analysis (LSA). The baseline theory for LSA is that by looking at the entire range of words chosen in a wide variety of texts, patterns will emerge in terms of word choice as well as word and document meaning. By investigating how participants' the choice of words changes throughout design discussions, LSA reveals the development of a coherent vocabulary. This allow examining whether a team manages to create a mutual understanding throughout a discussion, as well as to help identify triggers for development of such patterns. This temporal and quantitative approach has been a great inspiration for this work.

2.2 Data Visualisation for Improving Data Analysis

With the domestication of internet and digitalised devices, a new field of data visualisation has emerged in the last decade (Beck, Burch, Diehl, & Weiskopf, 2016). Hans Rosling and his Gapminder project is a perfect example hereof, and seeks to implement a fact-based worldview to anyone interested (Rosling, 2016). With Rosling's dynamic visualisations of demographic historical statistics he argues that the state of the world is not as bad as people might think. Rosling's claim that visualising data can change people's understanding of certain contexts are supported by studies on how the visualisation of data change the comprehensive performance of the user (Ware, 2009). Hence, there is reason to believe that researchers can gain new insights by presenting data sets in a graphical manner. In their review of state-of-the-art data visualisation tools Beck et al., (2016) conclude that data visualisation has been on the agenda since the early 90's, but the utilisation of the methods has turned into more *evaluating* character rather than only focussing on *method* or *application*.

2.3 Changing Group Interactions through Data Visualition

Visualising data concerned with language and dialogue patterns has also been an approach in the field of design research and educational research. This aims to investigate how the flow of discussion affects group performance (Avci & Aran, 2016; DiMicco et al., 2006; Ryu & Sandoval, 2015). Moreover, experiments have been made to investigate whether one could change group dynamics in a design discussion by providing participants with a visual live feedback of their contribution to a conversation in terms of speaking time and sound level (Bergstrom & Karahalios, 2007a; DiMicco, Pandolfo, & Bender, 2004). Bergstrom designed the so-called Conversation Clock that visualises the interaction patterns of up to four individuals, and provides participants with a *communal social mirror*. An interesting finding in their studies is that subjects looking at the visualisation of a group interaction could identify team roles and interactions without observing the actual group discussion.

Similarly, group performance was observed to correlate with speaking turns, speaking lengths (W. Dong, Lepri, Kim, Pianesi, & Pentland, 2012) and eye contact rate (Jayagopi, Sanchez-Cortes, Otsuka, Yamato, & Gatica-Perez, 2012). Avci & Aran (2016) developed a method for prediciting group performance. Their method take inputs such as ‘group speaking cues’, ‘group looking cues’, ‘personality trait statistics’, ‘Individual Speaking Statistics’, ‘Individual Performance Statistics’, ‘Group perception Statistics’, ‘Influence Cues’, ‘Hierachy Cues’ and ‘Frame-Based Features’. Based on case studies, they succeed in predicting the performance of groups when making decisions.

The field of visualisation of group interaction the goal is to actively affect or predict the discussion and speaking patterns by visualising the live flow of communication. For us, this means that it is plausible to hypothesise that by identifying modes of information sharing, one can gain insights on group behaviour.

3 Methodology

The strategy of this research has been to leverage as much quantitative information as possible from the DTRS11 transcripts and explore how visualising how one could visualise the data in a meaningful way for design observation researchers. Firstly, we explored and identified features from transcript data that are potentially relevant for visualisation. Then, a Python script was developed to extract selected features into a comma separated value (.csv-file) format. This file served as input to the network graph visualisation softwares Gephi and Cytoscape (Bastian et al., 2009; Smoot et al., 2011) enabling visualisation of the transcript. This approach led to the pilot study described in the following section. For evaluation of our first iteration, we conducted a qualitative interview with one of the researchers of the DTU group also working with the DTRS11 dataset (Christensen & Abildgaard, 2016). In this interview we compared reults and discussed the potential of mapping transcripts in a new and graphical way. From this evaluation we found some limitiations in our pilot study and with this in mind developed the final approach, Temporal Static Visualisation, based on new features from the transcripts.

4 Pilot Study involving Dynamic Visualisations

With the goal of developing a tool that could visualise conversation patterns, we started with a strategy of classical network analysis. For a pilot study we used the two softwares: Gephi and Cytoscape (Bastian et al., 2009; Smith, 2003; Smoot et al., 2011). Gephi and Cytoscape are both open source network analysis tools developed by researchers in the field of computer science

(Bastian et al., 2009; Smoot et al., 2011). The software allows researchers to explore and analyse all types of networks. The input is excel- or .csv-files designed by the researcher including relational information between nodes and edges. Both Gephi and Cytoscape allow animation of the development of a network over time. This was of our specific interest as it allowed us to get temporal insights in the group discussions in our data-set rather than getting one overall view. For the sake of consistency across figures the same data is used in all visualisations. Video 11 *Linking insights from co-creation to project*., segment 250 to 280 was selected for this purpose (Figure 1).

Figure 1 Video 11 *Linking insights from co-creation to project* was used as case illustration to secure consistency between the different types of developed visualisations

4.1 Creating data visualisations of speakerturns and amount of speach

By converting data from the trancsriptps into a csv.- file (see App A) we were able to both statically and dynamically show the communication flow of a discussion.

Static snapshots

Figure 3a shows the total aggregated speaker turns of Video 11 *Linking insights from co-creation to project*. Here, one see who is contributing to the conversation, and which speakers talk after each other. We see how Ewan and Amanda are contributing the most to the conversation, and show tendencies of talking amongst themselves. The graph also shows that Ewan is central to the discussion, as the most weighted edges connect to this node. Figure 3b show a 30 speaker turn sample of the same data, from (v11, 250) to (v11, 280). The previously described pattern of Ewan and Amanda dominating the conversation is even more apparent here. Static visualisations hence provide an overview of the speaker flow, but the actual sequence and quantity is still hidden.

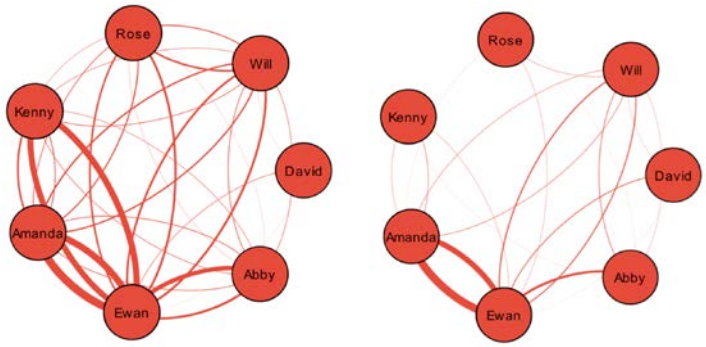

Figure 2a & 3b Static visualisations with weighted edges. Left: v11, all sessions.
Right: v11, s250-280

Dynamic Visualisation of Speaker Turns

To visualise the flow of conversation the dynamic possibilities of Cytoscape was utilised. Here one can visualise dynamic graphs of the temporal dimension in forms of videos. Seeing how speaker turns evolve turn by turn gave insights in team dynamics and showed the progression of the conversation. We approached dynamic visualisation by displaying ten speaker turns at the time. Figure 3 displays turns v11s250, s265, s280 each depicting the last the speaker turns as edges.

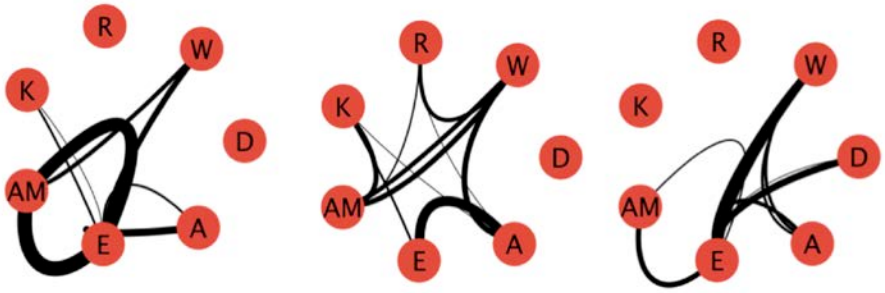

Figure 3: Dynamic network visualisation - Edges (v11, s250; v11, s265; v11, s280)

https://vimeo.com/172090367 (password dtrs11)

An alternative to displaying the last *n* edges in a dynamic visualisation is to adjust node sizes based on speaker activity. This gives a picture of how much each speaker is contributing to the conversation in the relevant time window. Figure 4 shows node sizes based on the subjects' activity in the last 20 speaker turns.

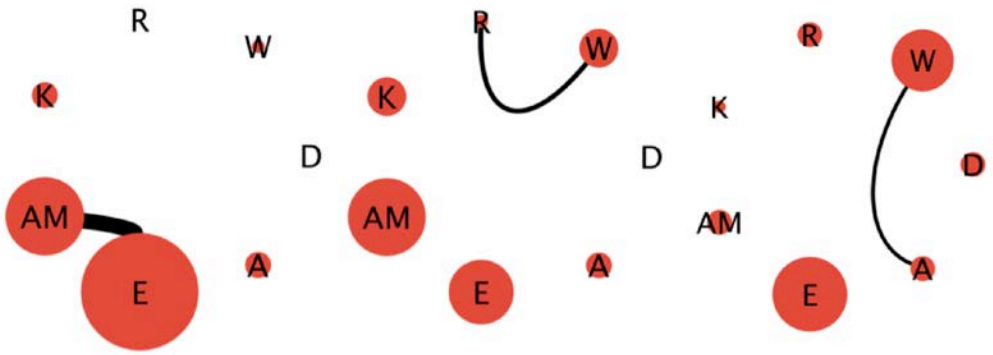

Figure 4 Dynamic network visualisation - Nodes (v11, s250; v11, s265; v11, s280)

https://vimeo.com/172090366 (password dtrs11)

4.2 Evaluation of the Automated Visualisation Approach

To evaluate our suggested approach we initiated a discussion with the DTU group. The evaluation consisted of comparing their manual coded schemes with our visualisations and a qualitative interview with one of the researchers discussing the value and potential of our pilot approach. Our claim regarding the potential of identifying patterns is supported by the work of the DTU group. One of their main findings was that there were statistically significant patterns in design discussions that could be identified through an approach like ours. However the research group mentioned they had coded for agreement expressions throughout the design group in a combination of speaking cues and body gestures such as nodding. This leads to the criticism that our visualisations lacked detail and did not yet help the researcher identify relevant sequences for further analysis. However, adding more contextual information such as use of artifacts or body language, the approach would have great potential according to the DTU researcher. He saw possibilities for identifying interesting passages in a video, and the potential of quantitatively comparing datasets across studies. Another critism was that the videos did not provide the intended overview of the dataset, which was one of the original intentions as well.

5 Developing the Temporal Static Visualisation

The development and evaluation of our pilot test failed to provide information on interesting sequences for further analysis. Even though the presentation of conversation flow has become more graphical, it still failed to provide useful holistic insights of the conversations. Hence, we developed a new approach seeking to do exactly this. By visualising the engagement of group members' participation throughout the group discussion along a time axis rather than an animate network we aimed at identifying different modes of information sharing such as monologues, dialogues and group discussions.

5.1 Mathmatical Explanation of TSV

Each individual participant p_i in the conversation is given a score denoting their contribution to the conversation over the last 20 speaker turns t. We define *Score* as a function of t_j and p_i (Equation 1). t_j is a given speaker turn (denoted ### in the supplied transcripts), e.g. t_j = 25. and p_i is a given participant, e.g. $p_i = p_k$ meaning Kenny. The calculated score is the product of participant activity and recent word contribution. *Activity* (Equation 2) is a ratio describing how

many times the given participant has contributed to the conversation in the last 20 turns in relation to the maximum contributions possible, *M*. Over the time period of 20 turns, the maximum possible speaker turns of one participant is ten - this would indicate the person speaking every second time in the sequence of 20 speaker turns. *Words* (Equation 3) is the ratio of words spoken of the given person over the last 20 turns in relation to the total number of words spoken in the same time period.

$$Score(t_j, p_i) = Activity(t_j, p_i) \times Words(t_j, p_i) \tag{1}$$

$$Activity(t_j, p_i) = \frac{1}{M} \sum_{t=t_j-19}^{t_j} (p_t = p_i \rightarrow 1), M = 10 \tag{2}$$

$$Words(t_j, p_i) = \frac{\sum_{t=t_j-19}^{t_j} (p_t = p_i \rightarrow \#words(t))}{\sum_{t=t_j-19}^{t_j} \#words(t)} \tag{3}$$

Plotting the *score* of each participant throughout the session of investigation results in a graphical representation as seen in Figure 5.

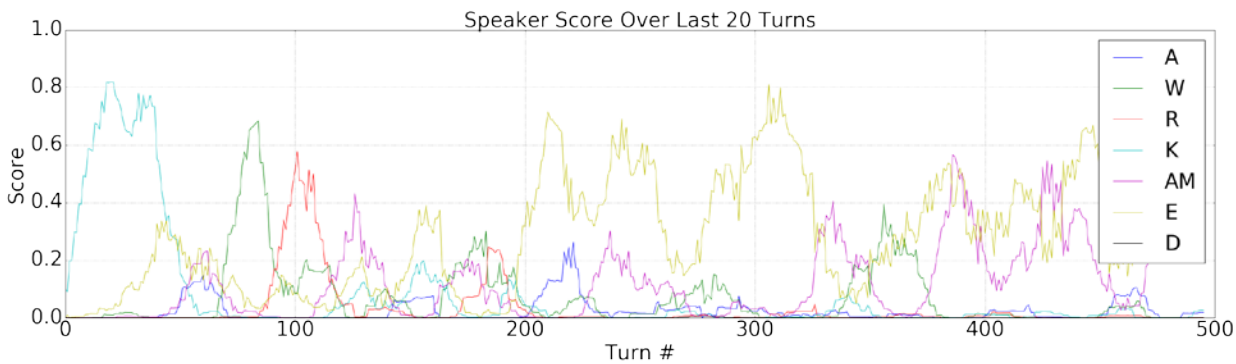

Figure 5 Speaker score calculated over last 20 turns.

Here, the contributions of each participant to the group conversation are visualised as a coloured curve, ranging between zero and one. E.g. in Figure 5 the turquoise curve indicates one participant speaking a lot in the beginning of the session and later becomes less involved. The yellow graph shows active participation throughout the conversation.

In order to identify modes of information sharing rather than individual contributions we calculated the variance of the score for each speaker turn to visualise the spread of activity among participants (Equation 4). This give an indication of periods of equal and more one-sided participation respectively.

$$Variance(t_j) = \frac{\sum_{p_i}^{p}(Score(t_j, p_i) - \overline{Score(t_j)})^2}{\#participants} \quad (4)$$

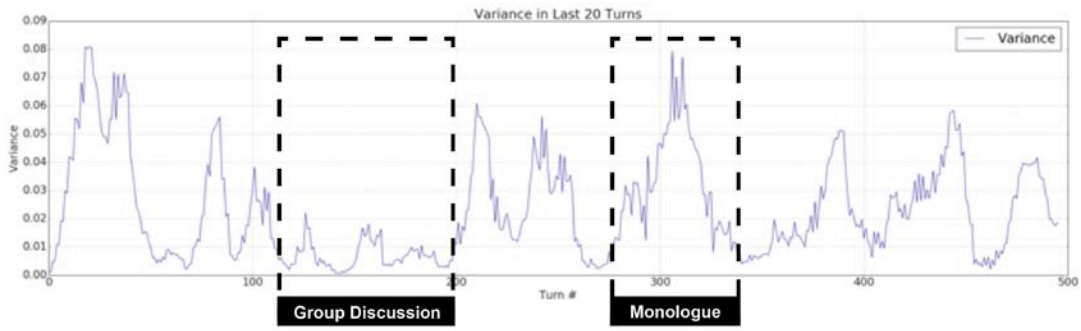

Figure 6 Variance of speaker score for each speaker turn. Group Discussion and Monologue is indicated in the dashed boxes.

In Figure 6 the variance of speaker scores throughout the conversation is shown. The variance calculated corresponds to the difference in activity among the participants for a given speaker turn. This indicate the type of information sharing happening throughout the conversation. When the variance is very low it several people are contributing to the discussion at the same time (*Group Discussion* in Figure 6). When the variance peaks around 0.1 it indicates one person is dominating the conversation (*Monologue* in Figure 6). Hence, the curve in Figure 6 provides the researcher with an insight into the occurences of group conversations, dialogues or monologues and the transition between these modes of information sharing.

5.2 Case Specific Findings

To show the application of TSV we have with the help of the visualisations been able to identify three sequences for further investigation. These are illustrated in Figure 7.

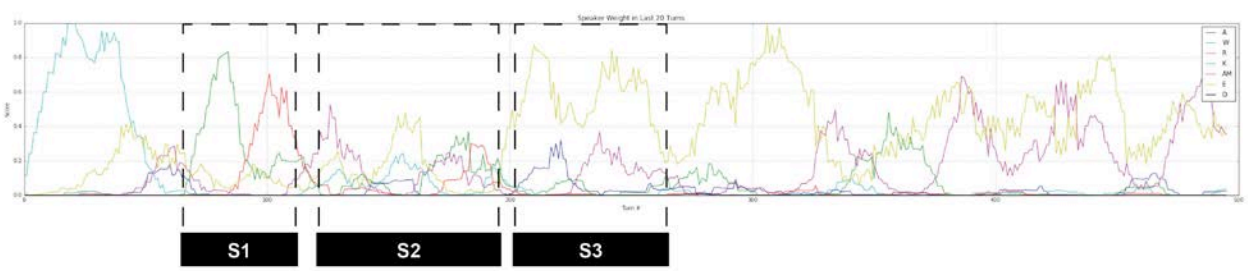

Figure 7 Illustration of three sequences (S1, S2 & S3) identified for further investigation

Sequence 1 (v11, s80-110) was chosen since we from the scores see how one person is contributing a lot to the overall conversation. Afterwards another person takes over. This could be an indication of a type of information sharing where participants spend longer time on sharing

a particular opinion on the same topic. Sequence 2 (v11, s110-200) was chosen as a period of equal participation based on scores and variance. This is the situation that most group dynamic researchers seek to create eg. with the conversation clock (Bergstrom & Karahalios, 2007b). Hence, it would be interesting to see the content of the discussion. Sequence 3 (v11, s360) was selected since the variance in this sequence is very high. This indicates that one participant is dominanting the conversation. Below we investigate the content of each sequence.

5.2.1 Sequence 1 (v11, s80-110) - Sharing Individual Concerns

Sequence 1 consists of two peaks of activity from Will and Rose respectively. These are the highest concentrations of activity from these participant throughout v11. Will begins this sequence by voicing concerns about the tone of the framing for the planned workshop. The concern is that the focus is currently on solving pains instead of enjoying good life, and is in his words *miserable*. Some discussion whether or not this is beneficial for the workshop ensues. Rose picks up on Will's concerns and continues the discussion of what the tone of the workshop should be like. She argues for a more positive tone more in line with the theme of good life. The conversation pattern in Sequence 1 can be described as concecutive large contributions of the participants.

5.2.2 Sequence 2 (v11, s110-200) - Group Discussion and Expressions of Agreement

Sequence 2 is identified as a period of relatively low variance in the group speaker activity. The start of this sequence is concerned with the notion of *pockets of time to enjoy good life*. The group is discussing whether they should focus on small pockets of time during work life, or the larger pockets of time the workshop participants can enjoy after they are 50 years old. The group seems to agree on placing the focus on smaller pockets of time in work life, what they call the "*pockets of the endurance*". Further, the group is discussing whether or not there will be any translation issues to Chinese and what sort of and the length of quotes from the earlier session to give the participants. This sequence is generally characterised by short expressions of agreement.

5.2.3 Sequence 3 (v11, s360) - Facilitating Coherence

Sequence 3 is a period of large calculated variance of the speaker score over the previous 20 turns. When looking at the scores, Ewan appears to dominate the conversation. The topics discussed here are first, how to make the workshop participants think about how to create trust in the company and second, to include the notion of sustainability in the discussion. This sequence contains elements where Ewan makes long explanations of concepts and is followed up by shorter expressions of agreement from other participants. In this way Ewan can be seen as a socalled *Intergroup Star* defined by Sonnenwald (1996). This communication role seeks to facilitate intergroup collaboration and alignment.

5.3 TSV for identifying different forms of information sharing

With the small qualitative investigation of sequence 1, 2 and 3 we show that TSV is able to single out different modes of information sharing. When looking into the actual conversation in each of the identified sequences it appears that group interactions are fundamentally different depending on the activity. We see indication of different team roles (Sonnenwald, 1996) as well as social group interaction such as agreement and expressing concerns (Bales, 1950). Based on this we find it reasonable to propose that these sequences could be of further importance for

design researchers since it is plausible to hypothesise that different types of group interaction result in different types of out-put in a design discussion. Hence with TSV we are able to identify sequences with higher reach potential fast, thus allowing the researcher to focus her resources better.

6. Discussion and Further work

With TSV, we have showcased a new way to explore video and corresponding transcripts in the pre-analysis phase. The strength lies in the ability to filter large amounts of information from video and transcript material into one static overview. TSV visualises what is otherwise invisible to the human eye - the actual communication flow and the historical participation of group members. Researchers can thus see how dialogue flows, enabling them to identify occurences of and transitions between monologue, dialogue and group discussion. This should allow researchers to gain a brief insight of the group dynamics in the team. We suggest to use these insights as input in identifying critical events the researcher should look further into. The final approach allow us to see how discussions develop over time at a glance as opposed to tools such as Leximancer and the network analysis approach described previously. Leximancer show a summary of the whole discussion, keeping the temporal dimension hidden, and the network approach still requires the researcher to watch long animations of the conversation. To summarise, TSV includes the temporal dimension while keeping the visualisation static. In this way it has rembelings to the semantic coherence analysis by Dong (2005).

A limitation of this work is the lack of contextual information in the visualisation. One of the drawbacks of our approach is exactly what Leximancer offers: content. Yet we believe that a combination of visualising participant activity with semantic analyses of different sequences in the graphs in Figure 5 would provide the researcher with an interesting starting point for tackling large data sets.

To illustrate this idea we extracted the text from the speaking cues connected to sequences S1-S6 in Figure 8 and used the software Leximancer to highlight the topics most frequent in the different sections. This can be seen in Figure 8.

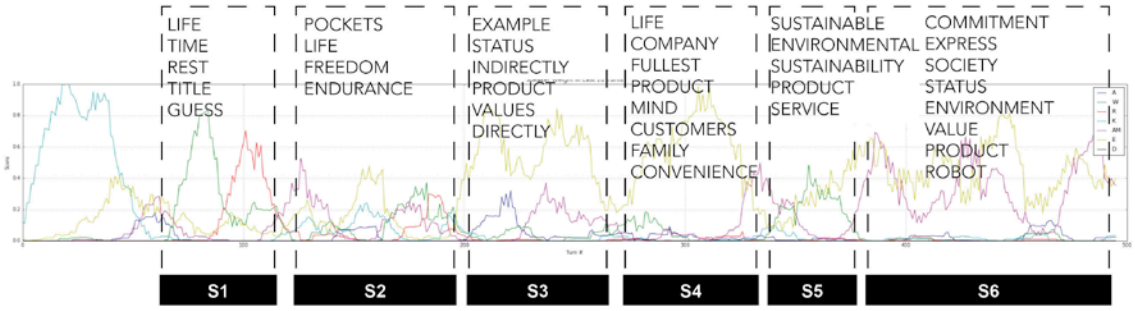

Figure 8 Illustration of how adding semantic content analysis to different sequences of the conversation could add more content related information to the temporal visualisation.

When adding keywords to the different passages of the conversation (Figure 8) we show how one easily can provide more semantic content from the transcripts. The approach does not yet provide any contextual information for the researcher studying more than group dynamics and conversation flow. This was one of the main criticisms given by the DTU researcher. As a design researcher studying more than "just" group dynamics, one would like information on interactions (e.g. artefacts) as well as body language and directional attention from the participants. This want is supported by the fact that several of the reviewed sources both in the field of design research and group dynamic visualisation include more than only speaking queues (Avci & Aran, 2016; DiMicco et al., 2006; Sonalkar et al., 2013). Hence, we face the challenge of how to include more details in our graphical representations.

Looking at the 13 interactions defined by Sonalkar et al. (2013) we see that at least some of them could be captured in a transcript. This concerns questions (by a "?"), interruptions (coded a "-"), yes and (by "yes and") and humour (by "[laughing]"). Also since the LSA method of Dong (2005) already include semantic development over time it would be an obvious candidate to combine with our temporal visualisations.

7 Conclusion

This paper presents the development the Temporal Static Transcript Visualisation - an approach for pre-analysis of video material captured from observational studies of group dynamics. By utilising the meta-data of already written trancripts we map the flow of conversation throughout the video by visualising participation in terms of amount of words said and how often a person talks. Moreover we visualise how many group members who are taking an active part of the conversation over time. In this way we are able to identify different types of dialogue patterns eg. monologue, dialogue and group discussion. With TSV we help qualitative researchers to gain a faster overview of the overall conversation flow of a studied group collaboration.

As a pilot study we visualised the conversation dynamically with the help of the software Cytoscape. To evaluate our suggested approach, we conducted an interview with a fellow research group also working with the dataset from DTRS11 (Gonçalves & Cash, 2016). Firstly, we compared their manual coding with our visualisations. Further, a qualitative interview discussing the value and potential of the approach was conducted.

Through this evaluation we learned that the pilot test was yet too "dynamic" and failed at actually helping the researcher identifying different forms of information sharing types which was the original idea. Hence we developed the Temporal Static transcript Visualisations (TSV) that over time graphically presents group individual participation as well as the overall group activity. With the visualisation we were now able to identify sequences of and transitions between monologues, dialogue and group discussions. We propose to use this as a starting point for analysing bigger set of group discussion obervations. This approach was illustrated by picking out three different sequences in v11 and qualitative analysing them. It was confirmed that the sequences were fundamentally different and also included phenomena as team roles or social group interaction types. Hence we argue that TSV has a relevance in the Design Research community opening up for quantitative analysis among data sets.

However we have some concerns on how TSV can become even more valuable for design researchers and not only group dynamic researchers. There are some limitations on only providing an overview of communication flow since such representation might be too simple a representation for a design researcher to retrieve valuable insights. Adding details such as the use

of artefacts or body language could make our approach into an interesting tool not only for identifying moments of interests, but to actually create comparable output and allow for analysis between case studies. Our further work will focus on how we can combine semantic methods with our algorithm and allow the visualisation of more information to provide content based impressions. In this work we contribute with first attempts to make the process of analysing design observation data less time consuming in the individual cases and comparable across cases.

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Appendix A

4.1 From Transcript to .csv-file (Step 2-3 in Figure 1)

In the case of DTRS11 we exploit the fact that the conversations in the source material is already transcribed with numbered speaker turns, initials, and content (Table 1). This allowed us to extract chronological development of conversation (last and future speaker) and number of words mentioned in each speaker turn as well.

Table 1. Supplied transcript, stripped of superfluous data

###	initial	TRANSCRIPTION
255	AM	Optimize them.
256	K	Mhm.
257	E	Yeah, and the meaning is something maybe you find in the end, but you maybe have endured a long life, to find that in the end.
258	A	But I guess, living life to the fullest is, what you're also saying, to doing- enjoying life and doing what matters to them and what they want to do. All of that is I think what we need with living-
259	W	Yeah, it captures that- probably:- probably Amanda's point is that nobody's saying living life to the fullest has more the connotation of just (.) whole out enjoyment.

Based on the sequential format of the data, where each row represents one speaker turn, a Python algorithm has been written to parse transcripts into a directed network graph format (See Appendix 1). Here, nodes represent individual speakers, and edges represent a connection between the current speaker and the next speaker. In Table 2, the first data row shows an edge E-W going from node E (Ewan) to node W (Will). The sequence of edges is denoted by the start time column, corresponding to the original speaker turns in the transcript. In addition to translating speaker turns into nodes and edges, the number of times a subject has spoken in the n last turns (Activity), and how many words they have uttered (#words) were extracted. To reduce the impact of #words on the weight calculation of edges, which would affect the visualization dramatically, the natural logarithm of #words were calculated and applied in its place. These are initial suggestions of features used in our first iteration of our suggested approach. All data processing has been done in Python, using the Graphlab Create (Low et al., 2014) and Numpy (Van Der Walt, Colbert, & Varoquaux, 2011) software packages.

Table 2. Network graph format, features calculated over the last 20 turns.

Source	Target	Start Time	Stop Time	Activity	Weight	#words	log_words
AM	K	255	275	1	1.10	2	1.10
K	E	256	276	2	1.39	1	0.69
E	A	257	277	3	9.89	26	3.30
A	W	258	278	1	3.74	41	3.74
W	AM	259	279	1	3.37	28	3.37

Requirements for Dynamic Visualisation

For dynamic visualisation timestamp data needs additional preparation and changes. Cytoscape and the plugin DynNetwork were chosen for visualising dynamic graphs, and the dynamic graph data is processed according to the software's input requirements. Table 3 presents node data pre-processed for use in Cytoscape. Each node is represented by a unique numerical node id along with a label. Type and fill are pre-defined values determining the shape and colour of each node. The size variable determines the size of each node, and can be calculated based on e.g. the subject's participation in the previous n speaker turns. With the starting time of each edge from Table 2, we can define how long a node is active in the visualisation based on user input.

Table 3. Nodes in dynamic graph format

NodeId	Label	type	fill	size	StartTime	EndTime
1	A	ELLIPSE	#E74C3C	20	255	256
2	W	ELLIPSE	#E74C3C	20	255	256
3	R	ELLIPSE	#E74C3C	20	255	256
4	K	ELLIPSE	#E74C3C	20	255	256
5	AM	ELLIPSE	#E74C3C	20	255	256
6	E	ELLIPSE	#E74C3C	20	255	256
7	D	ELLIPSE	#E74C3C	20	255	256

Table 4 shows the corresponding edge data to the nodes described in Table 3. For Cytoscape to recognise the data as edges. The two first columns need to be named FromId and ToId. The weight parameter is taken from Table 2 and is used for edge width when visualising the data. Fill governs the colour of edges, and Start/End time serves the same function as described above for nodes.

Table 4. Edges in dynamic graph format

FromId	ToId	weight	fill	StartTime	EndTime
5	4	0.79	#000000	255	265
4	6	1.07	#000000	256	266
6	1	4.59	#000000	257	267
1	2	2.71	#000000	258	268
2	5	2.51	#000000	259	269
5	4	0.79	#000000	255	265
4	6	1.07	#000000	256	266